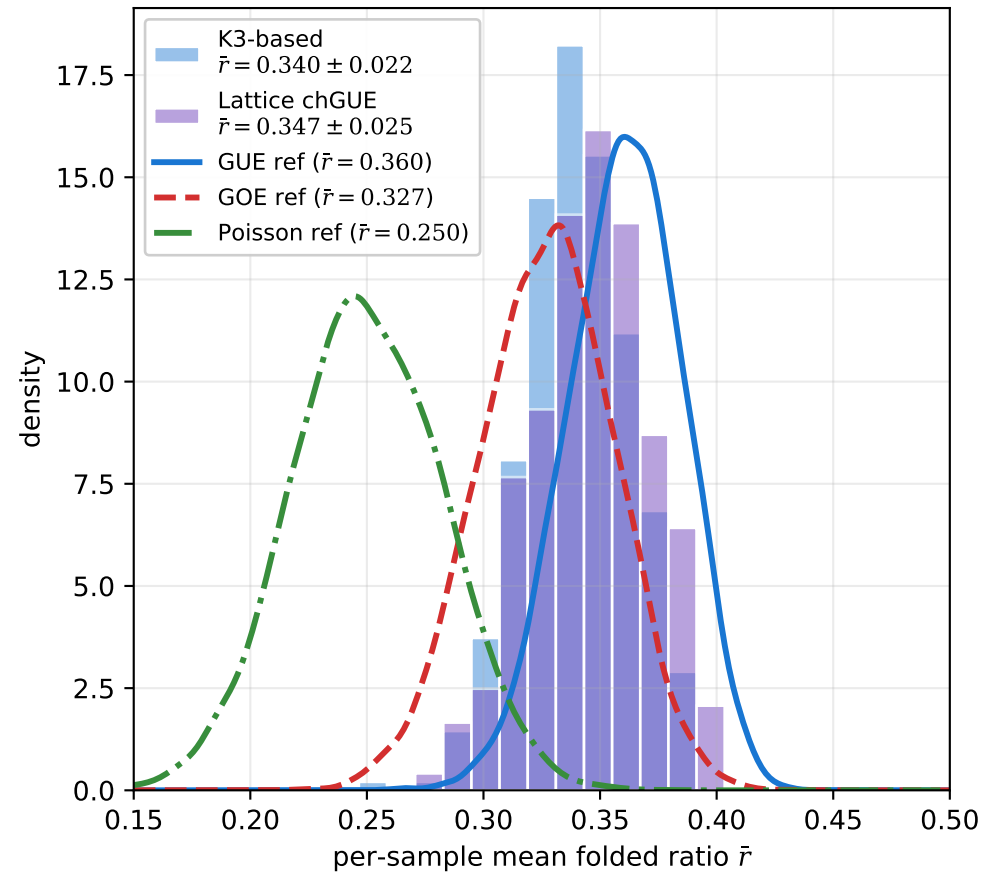


First-principles lattice-QCD replacement: $Q_{K3} \rightarrow \gamma_5 D_W^{(22)}$ (chGUE, $N_f = 6$, $\nu = 0$). Same $N = 28$, stronger GUE signal at smaller κ_T .

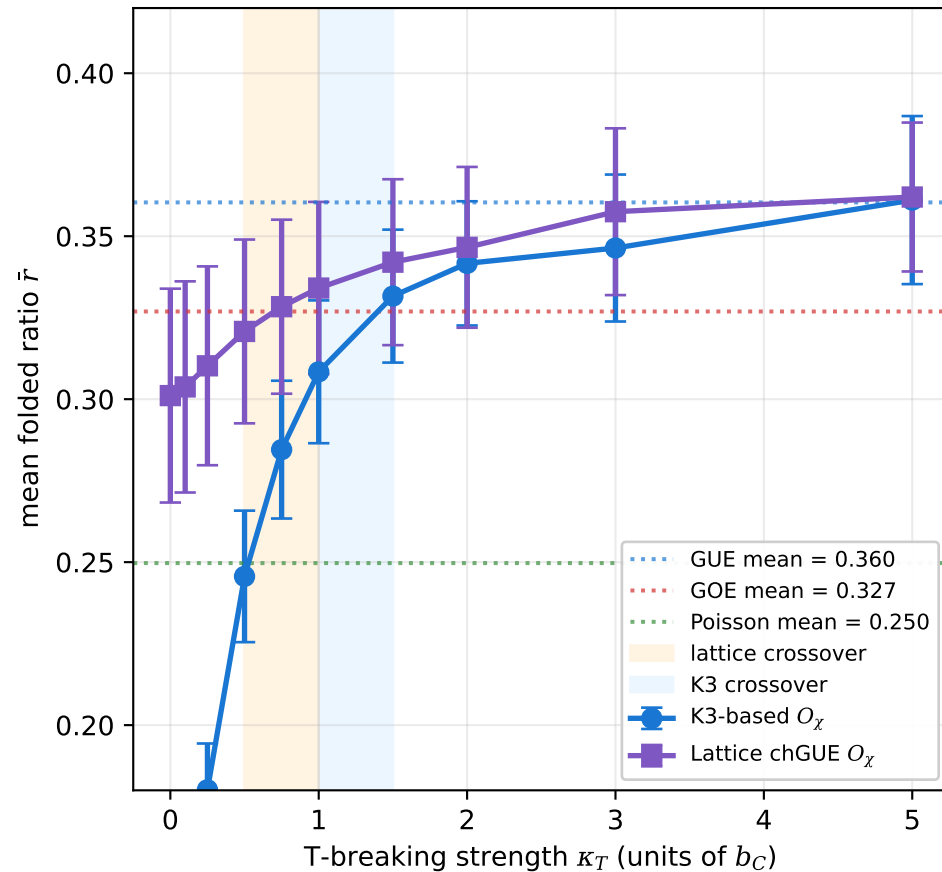
O_χ at $\kappa_T = 2$ (physical QCD vacuum)

K3 topology vs first-principles chGUE



Lattice reaches GUE at smaller κ_T

(chGUE is already in the T-broken class)



First-principles lattice chGUE beats K3

at every κ_T -- same physics, stronger signal

